

# Md. Ashraful Hossen Akash

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## RESEARCH INTERESTS

Computer Vision, Large Language Model, Natural Language Processing.

## EDUCATION

**B.Sc. in Computer Science & Engineering**

Mar 2022 - Jun 2026

Rajshahi University of Engineering & Technology

**Result (CGPA): 3.17/4.00 (3.61 is last 3 semesters)**

## PUBLICATIONS

- **Md. Ashraful Hossen Akash**, Shyla Afroge, Md. Asif Sajeed, Md Barkullah, Abul Khayer. “An Explainable Deep Learning Framework for Multi-Class Alzheimer’s Diagnosis with Multi-Scale CNN.” **(Accepted)**  
*2nd Conf. on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2026)*. [\[ResearchGate\]](#)
- Md. Asif Sajeed, Shyla Afroge, **Md. Ashraful Hossen Akash**, Abul Khayer, Abdullah Al Mamun. “Memory-Efficient Deep Learning Architectures for Water Body Segmentation in Satellite Imagery: A Comparative Analysis and Ensemble.” **(Accepted)**  
*28th International Conference on Computer and Information Technology (ICCIT 2025)*. [\[ResearchGate\]](#)
- Abul Khayer, Shyla Afroge, Abdullah Al Mamun, Md. Asif Sajeed **Md. Ashraful Hossen Akash**, et al. “Transformer-Based Multi-Label Disease Prediction from Radiology Reports.” **(Accepted)**  
*28th International Conference on Computer and Information Technology (ICCIT 2025)*. [\[ResearchGate\]](#)

## EXPERIENCE

**Research Assistant**

May 2024 – Present

Supervised by Shyla Afroge

*CSE, RUET*

- Conducted research in deep learning for medical imaging, satellite imagery, and clinical text analysis, including data preprocessing, model development, and experimental evaluation.
- Contributed to three accepted international conference papers (ICCIT 2025, QPAIN 2026) involving multi-scale CNN architectures, transformer-based NLP models, and ensemble deep learning methods.

**Software Engineer Intern**

Feb 2025 – May 2025

Code Studio Software Company

*Rajshahi, Bangladesh*

- Gained experience with software development workflows and the Software Development Life Cycle (SDLC); designed relational database schemas and wrote SQL queries (CRUD, JOINS, aggregates).
- Built a full-stack CRUD/Blog application using Node.js, Django, React, and Next.js with MySQL, MongoDB, and PostgreSQL; implemented REST APIs, authentication, middleware, and rate limiting.

**EDGE Training Program (PHP – Laravel)**

May 2024 – Feb 2025

Bangladesh Computer Council (ICT Division)

*CSE, RUET*

- Completed an 8-month professional training under the “Enhancing Digital Government and Economy (EDGE)” project, gaining hands-on experience in backend development using PHP and Laravel.
- Developed MVC-based CRUD web applications and implemented database integration, authentication systems, and RESTful APIs following modern web development practices.

**Private Tutor (Mathematics, Physics, Programming)**

Mar 2022 – Present

- Tutored secondary and higher-secondary students in Mathematics, Physics, ICT, C programming, HTML, and database fundamentals.
- Guided students through problem-solving techniques, exam preparation, and foundational programming.

## COMPETITIONS

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### Kaggle Competitions

- **Santa 2025 - Christmas Tree Packing** — Rank: **425/3357 (Top 13%)**.
- **Predicting Student Test Scores** — Rank: **96/4317 (Top 3%)**.
- Applied feature engineering, model training, and ensemble techniques to improve leaderboard performance on real-world tabular datasets.
- Kaggle profile: [Ashraful](#)

### Codeforces Competitions

- Active participant in Codeforces rated contests (Div. 2, Div. 3, Div. 4). Current rating: **995**.
- Solved algorithmic problems involving arrays, greedy techniques, data structures.
- Codeforces profile: [ash\\_ra\\_ful](#)

## PROJECTS

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### AlzDetect-Ai: Explainable Alzheimer's Disease Classification System

Developed an AI-powered system for multi-class Alzheimer's disease diagnosis from brain MRI scans using a custom multi-scale CNN and an ensemble of ResNet50 and Xception models.

- Trained and evaluated the model on 33,000+ MRI images, achieving strong multi-class performance across four disease stages.
- Applied explainable AI methods (Grad-CAM, Integrated Gradients, LIME) and deployed the model as a lightweight Streamlit web application for real-time inference.

*Technologies: Python, TensorFlow/Keras, CNNs, Transfer Learning, XAI, Streamlit*

[\[GitHub\]](#)   [\[Live Demo\]](#)

### Python Basics: Step-by-Step Learning Repository

Designed and maintained a structured Python learning repository for beginners and early-stage CSE students, covering core programming concepts through progressive, hands-on modules.

- Organized Python fundamentals into 18 sequential modules covering variables, data types, conditionals, loops, strings, collections, and problem-solving with practical examples.
- Developed interactive mini-programs (calculator, shopping cart, countdown timer) to reinforce learning and support self-study, exam preparation, and introductory programming practice.

*Technologies: Python*

[\[GitHub\]](#)   [\[Learning Website\]](#)

### Restaurant Management Web Application

Built a Laravel-based full-stack web application with MVC architecture, database integration, and responsive UI using Bootstrap.

[\[GitHub\]](#)

## SKILLS

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|----------------------------------|-------------------------------------------------------------|
| <b>Machine Learning &amp; DL</b> | CNNs, Transformers, Explainable AI (XAI), Transfer Learning |
| <b>Programming Languages</b>     | Python, C, PHP, JavaScript                                  |
| <b>Data &amp; Databases</b>      | MySQL, PostgreSQL, MongoDB, SQL                             |
| <b>Web Technologies</b>          | HTML, CSS, Bootstrap, Laravel, React, Node.js               |
| <b>Tools &amp; Platforms</b>     | Git, GitHub, Kaggle, Codeforces, Linux                      |

## REFERENCES

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### Shyla Afroge

Associate Professor, Department of Computer Science & Engineering  
Rajshahi University of Engineering & Technology (RUET)